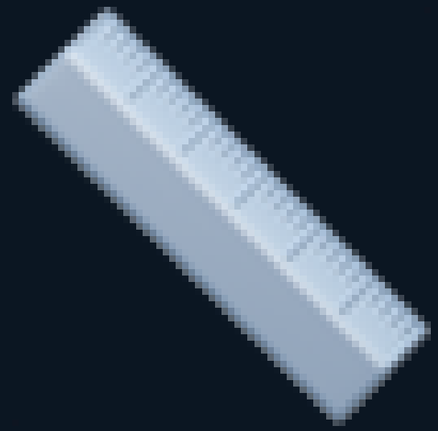
WEEK 3 · DAY 2

Control Charts and Residuals



Know what “normal” looks like. Then anything outside it jumps out at you, like a wrong price at the market. 🍅

NIGCOMSAT
AFRICAN ROOTED GLOBALLY POSITIONED

Publica Academy

YOU ALREADY DO THIS

Mama Put knows her tomato prices 🍅

Two weeks at the market and she knows the **normal range**. One day the basket costs way more.

She doesn't need a rulebook. She knows **normal**, so she spots **strange**.

That's a control chart, running in her head.



PART 1

Control charts

A centre line, two limits, and the discipline to leave normal alone.

ANATOMY

Three lines, all learned from a quiet stretch

—

Centre line

the **mean** of a known-good **reference period**

⬆️

UCL

upper control limit = centre **+ 3σ**

⬇️

LCL

lower control limit = centre **− 3σ**

❌ **Not the min and max ever seen**

One bad day would stretch them forever.

❌ **Not the engineering design limits**

Those say “broken”. Control limits say “unusual”.

STEP 1 · PICK A QUIET REFERENCE PERIOD

1-15 August: sunlit, no eclipse yet, nothing strange

360
hours of solar-array current

27.87
centre line (amps)

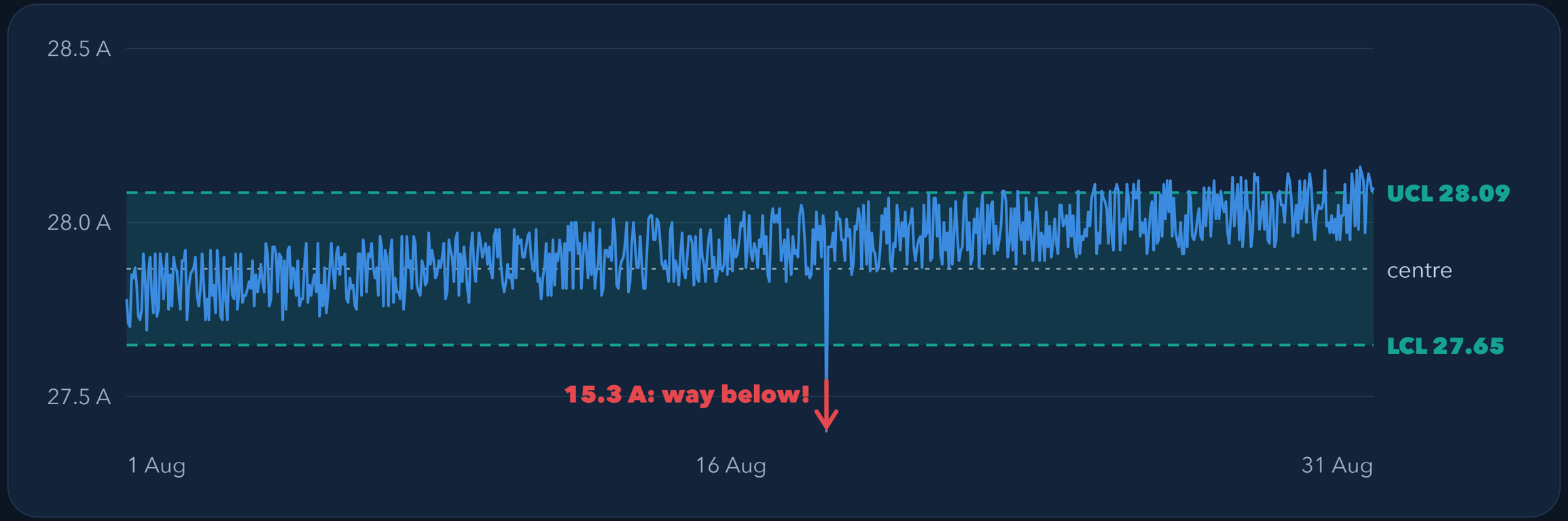
0.073
σ, the normal wobble (amps)

27.65–28.09
control limits (amps)

```
ref = bus.loc["2025-08-01":"2025-08-15", "solar_array_current_a"]
centre, sigma = ref.mean(), ref.std()
ucl, lcl = centre + 3 * sigma, centre - 3 * sigma
```

STEP 2 · WATCH NEW READINGS AGAINST THE BAND

August: one hour falls straight through the floor



18 Aug, 09:00: the array gave **15.34 A** in full sun. About half its normal output. Remember NigComSat-1 in 2008? 🛰️

WHY 3Σ AND NOT 2Σ?

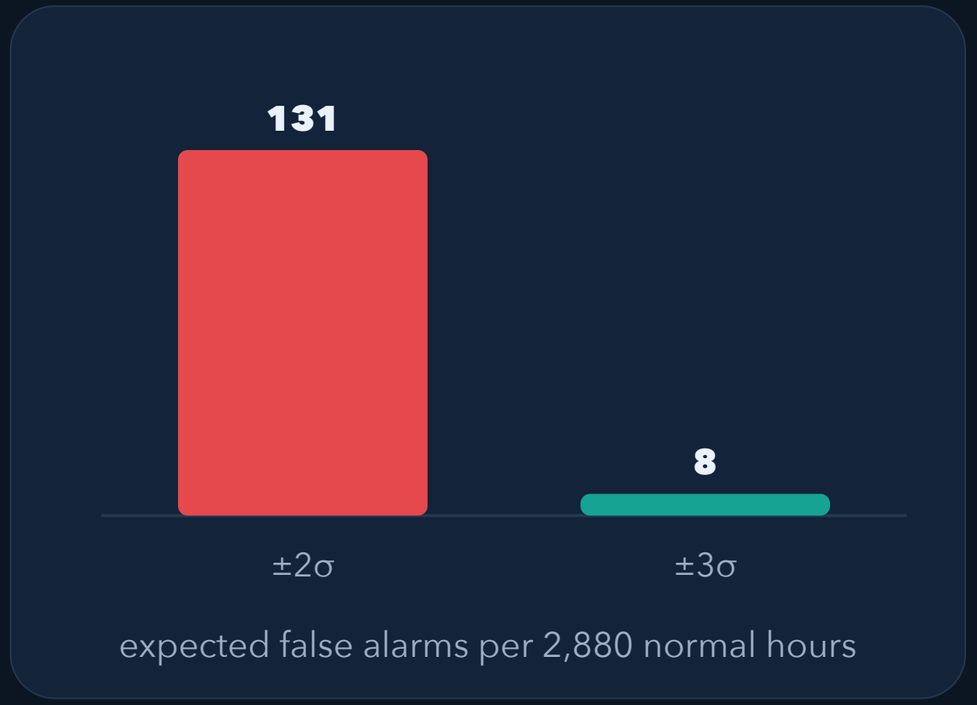
Every alarm costs attention

On perfectly normal data, how many hours land outside the limits by pure chance in **2,880 hours** (120 days)?

±2σ → 4.6% of hours → **≈ 131 false alarms** 😩

±3σ → 0.27% of hours → **≈ 8 false alarms** 😌

On our bus data with rolling limits: ±2σ fires 31 times, ±3σ fires 5.



expected false alarms per 2,880 normal hours

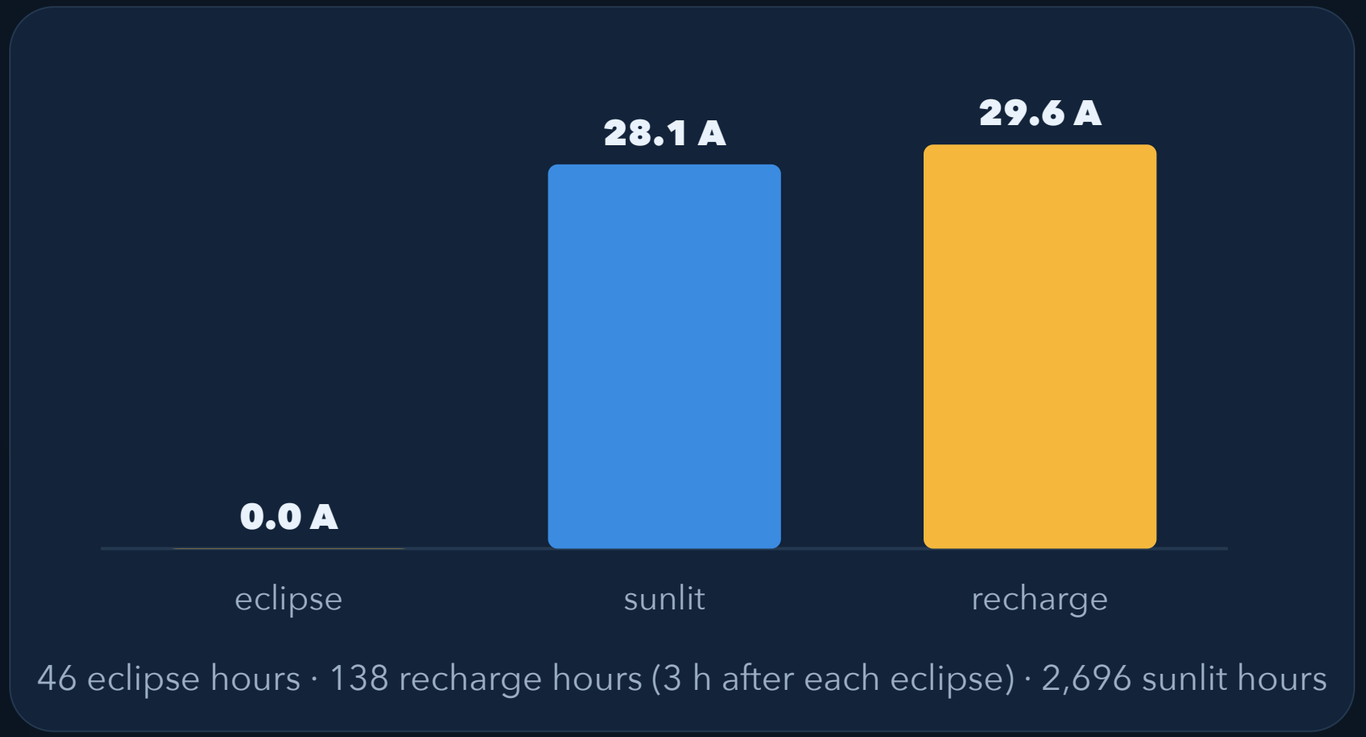
OPERATING STATES

One satellite, three moods

In eclipse the array gives ~0 A. Just after eclipse it works **extra** to recharge the battery.

Rule: separate limits for each state

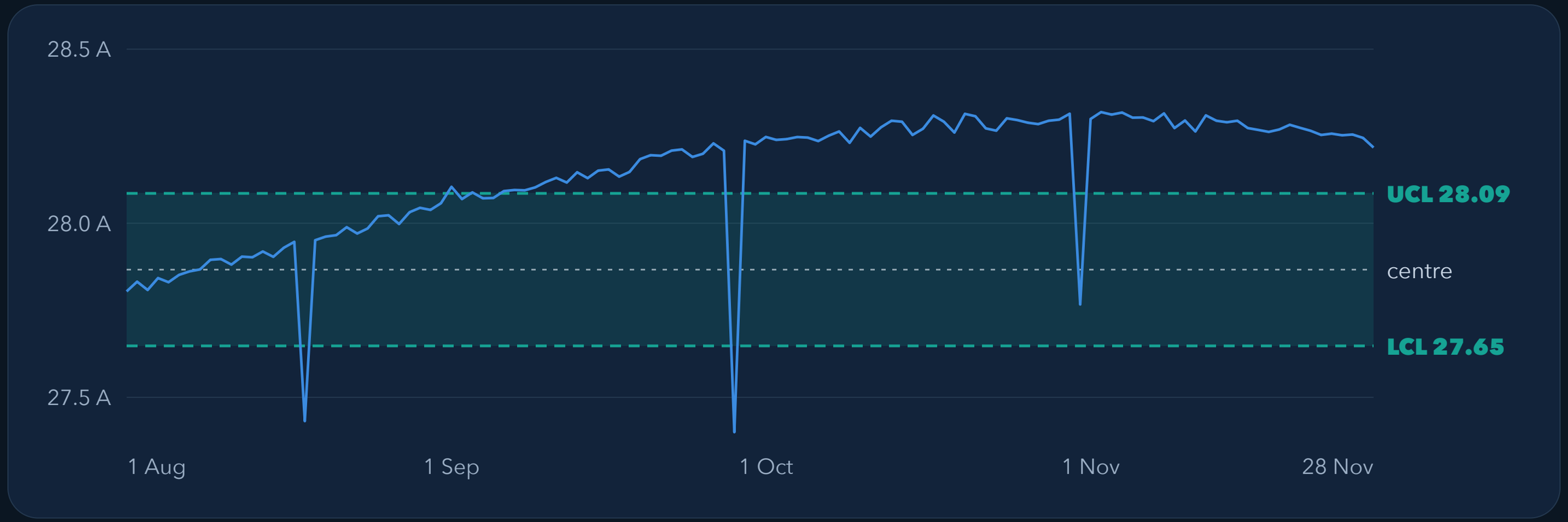
Mixed, σ grows from **0.073** to **0.594 A** and nothing fires. Don't delete hours or drop to 2σ.



46 eclipse hours · 138 recharge hours (3 h after each eclipse) · 2,696 sunlit hours

THE CATCH

A fixed reference goes stale: “normal” slowly moves



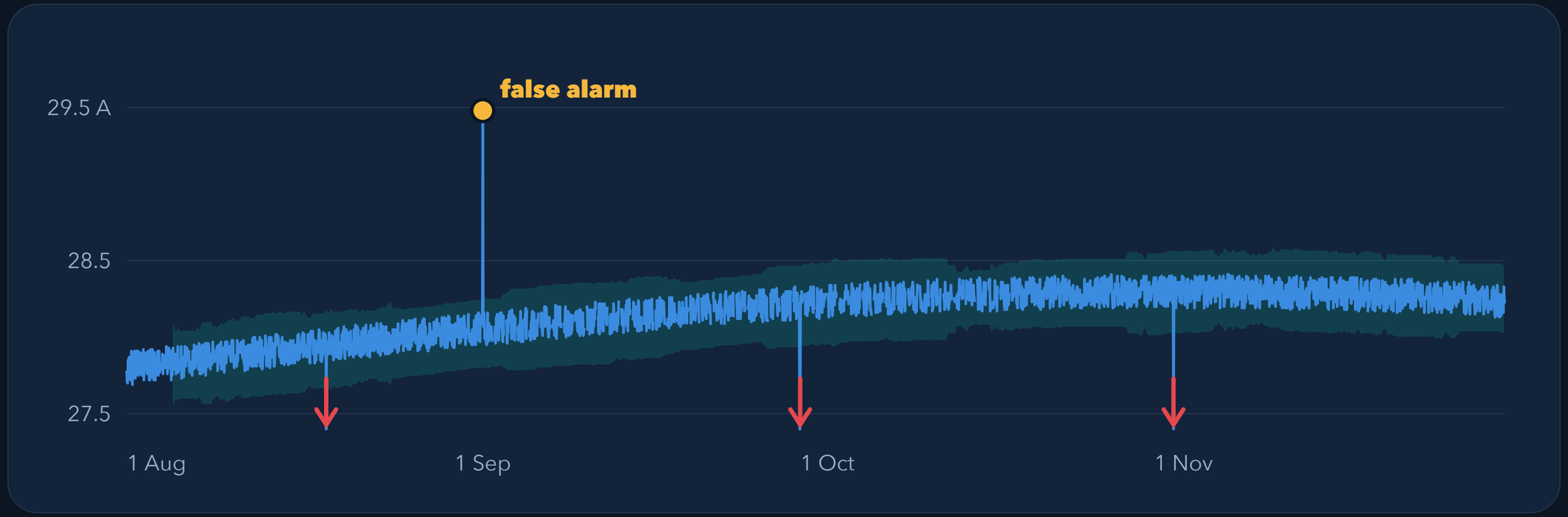
1,861 of 2,696 sunlit hours outside the August limits

daily average of sunlit current

sun angle slowly raises the output through the season

THE FIX · ROLLING LIMITS

Learn “normal” from the last 7 days, every hour



band = rolling median ± 3 robust σ (trailing, sunlit hours only)

5 flags = 4 dropout hours + 1 false alarm

WHY MEDIAN AND MAD?

One bad hour should not move “normal”

Mean + standard deviation

Right after the first 28 Sep dropout hour, the rolling σ jumps to

1.08 A

One wild point poisoned the ruler.

Median + MAD (robust)

Same hour, the robust σ is still

0.09 A

A few extreme points can't move it.

MAD × 1.4826 = same size as σ for normal data, so “3 robust σ” means what you expect.

COMMON CAUSE VS SPECIAL CAUSE

Don't touch normal wobble 🙅🏾

Common cause: the everyday wobble inside the band. Leave it.

Special cause: outside the band, or a strange pattern. Investigate.

Tampering = reacting to common-cause wobble as if it were special. It makes things **worse**.



The driver who jams the brake for every small bump on the expressway

39%

more wobble when you “correct” every reading (simulation)

RUN RULES

Not just “one point outside”

Rule 1: one point beyond 3σ

Catches **spikes** and dropouts.



Rule 2: nine in a row on one side

Catches **level shifts**: nothing extreme, just “stuck high”.



illustration

Keep Rule 2 in your pocket. Tomorrow's slow-creep detector (CUSUM) is its big brother. 👀

PART 2

Residuals 🔍

Stop asking “is this number big?”. Ask “is this number **surprising**?”

THE IDEA

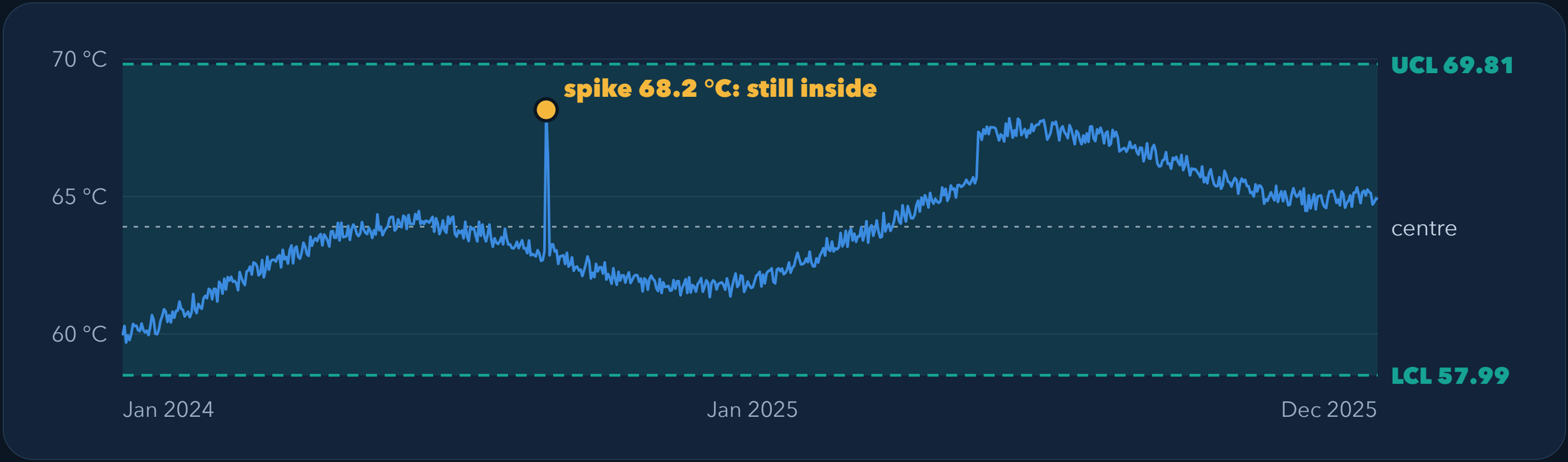
Residual = what happened − what you expected



Mama Put expects tomatoes to be dear when they're scarce. High price in the scarce season? **Expected.** High price when the market is flooded? **Surprise.**

AMPLIFIER (TWTA) TEMPERATURE · TWO YEARS, DAILY

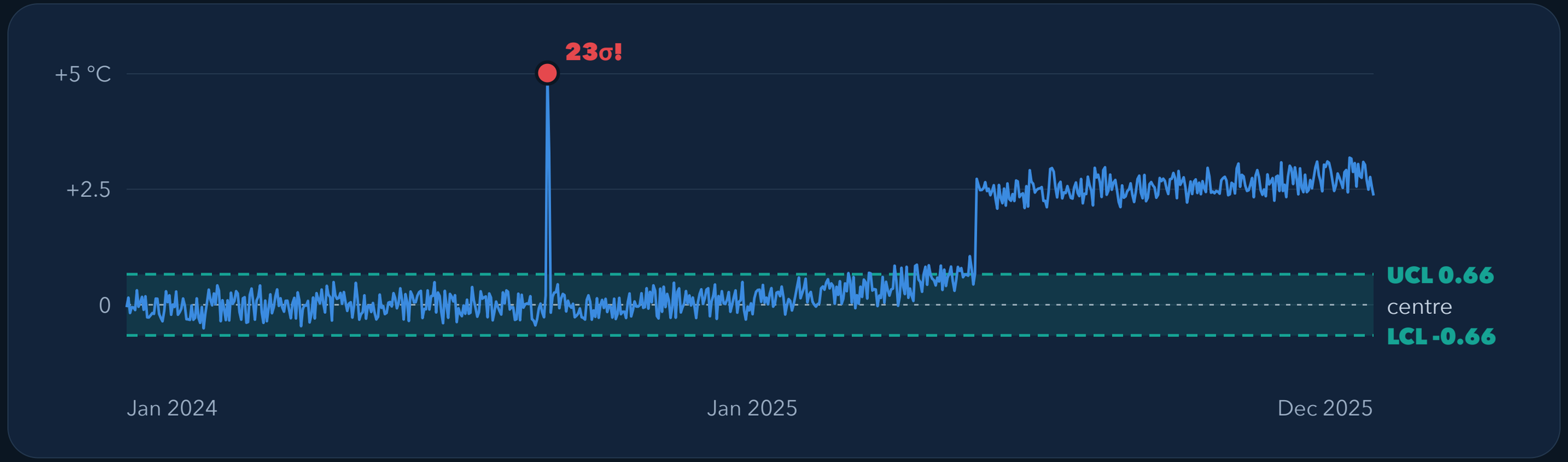
On the raw chart, the spike hides inside the band



Temperature also rises with **drive level** and the **sun angle**, so the raw band must be wide (σ = 1.97 °C). The 68.2 °C day never crosses 69.8 °C.

EXPECTED = A RECIPE LEARNED FROM 2024

temp ≈ 49.5 + 0.17 × drive + sun-angle wave



σ 0.22 °C (raw: 1.97)

spike +5.0 °C = 23σ

after 12 May 2025: 233 of 233 days flagged

IS YOUR RESIDUAL CLEAN?

Three health checks before you trust it

⚖️

Mean near zero

the recipe isn't biased high or low

🔗

No autocorrelation left

today's surprise doesn't predict tomorrow's

📏

Constant spread

same wobble in every part of the year

	mean	lag-1 autocorr	spread Jan–Jun / Jul–Dec
Recipe fitted on all of 2024	0.00	0.33	0.22 / 0.49
Refit without the 2 spike days	0.00	0.06	0.22 / 0.22

Two bad days in the training data spoiled the recipe. “Values all above zero” is **not** a health check: half should be below!

CAREFUL WITH CENTRED WINDOWS

A centred window swallows part of the spike

Trailing 7-day median

expected = the week **before**

+5.3 °C

residual on 3 Sep 2024

Centred 7-day mean

expected includes the spike itself (and the future)

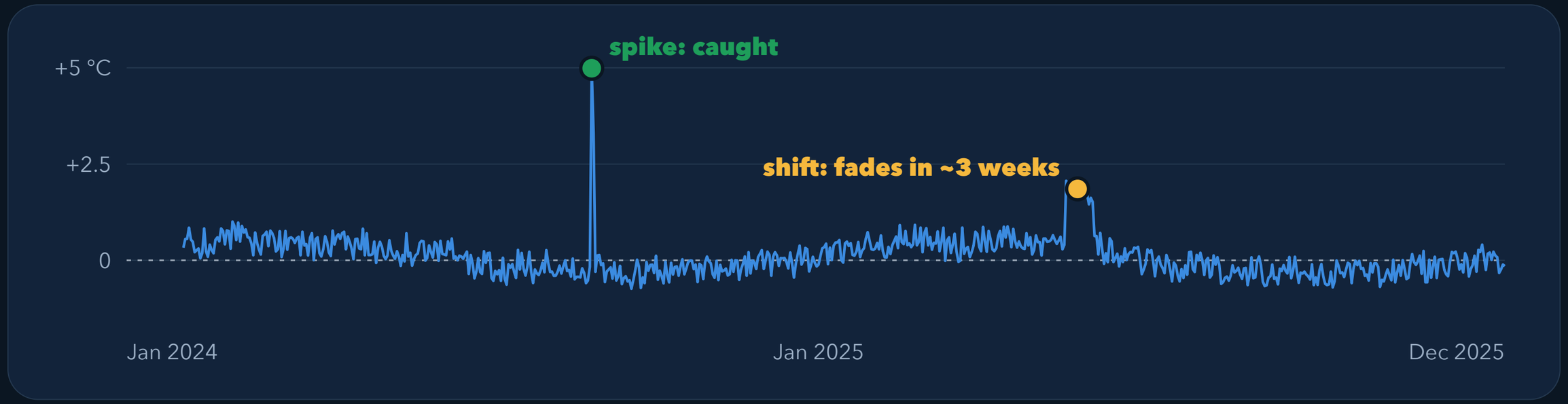
+4.0 °C

residual on 3 Sep 2024

The spike is **inside** its own window, so it drags “expected” up and hides part of itself. Wider windows hide less, but it's still cheating with the future. Trailing only. 🔮🚫

THREE SHAPES OF TROUBLE

A rolling “expected” follows slow change… so it misses it



💥 **Spike**: caught (11 robust σ)

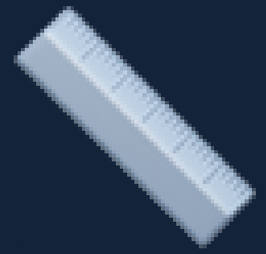
🪜 **Level shift**: **0** flags after 12 May

🦕 **Slow drift**: never seen

YOUR TURN · 60 MINUTES

Open the W3D2 exercise 👩🏾‍💻

- **A** Build a control chart for the solar array, and watch it go stale
- **B** Rolling robust limits, then residuals on the amplifier
- 🌶️ Bonus: prove tampering makes things worse



14 checks to pass

File → Save a copy in Drive first!
Run STEP 0, then top to bottom.

Keep the **rule book** open next to you.

TODAY IN 3 LINES

What you now know

📏 **Control chart:** centre ± 3σ from a quiet reference, **one set of limits per operating state**, refreshed so it doesn't go stale.

🛡️ **Robust:** median + MAD × 1.4826, so one wild hour can't move “normal”. Don't tamper with common-cause wobble.

🔍 **Residual** = observed − expected. Check it's clean. Great for spikes, weak on shifts, useless on slow drift.

Next 👉 **The Slow Creep: Drift and CUSUM** 🏚️